1. Executive summary

What was investigated

The brief's product scope (accounts, earning tasks, rewards, wallet, withdrawals, referrals, engagement and support) was turned into a complete analytics foundation: a data-readiness audit, a 38-event tracking plan, a 100-field data dictionary, a logical data model, 46 KPIs, quality controls, dashboard specifications, an engineering handoff, a pipeline, governance rules, a change process and a prioritised backlog. Two simulated workbooks were then profiled and audited with reusable, tested code.

What was discovered

- **Two Users tables contradict each other.** Both files use the same 1,000 user IDs (VLUSR000001 to VLUSR001000), yet names, emails and registration times differ for all 1,000 users. One table must be declared the user master before any join.
- **The supplied pack is incomplete.** Rewards, Wallet, Withdrawals and Referrals tables, which the brief lists, were not supplied. The behaviour file contains users, sessions, 35,704 events, daily activity, task activity and milestones only.
- **Structure is sound but not production-ready.** No orphan user or session references exist, but task attempts carry no reward amount or unit, the task catalogue is missing (every task_id maps to several categories), and stored counts do not reconcile (events_in_session wrong for 7,537 of 7,913 sessions; 423 daily rows with more completed than total tasks).
- **The messy sheet behaves as designed.** Automated checks found 8 duplicate user rows, 9 missing required values, 1 malformed date and 2 conflicting registration times; the event audit sheet holds 10 duplicate events, 4 trailing-space statuses and 5 inconsistent feature names.
- **Personal data is over-collected.** Full name, email, age, gender and city are present for every user, contrary to the brief's data-minimisation rule.

What is currently unknown

Whether any event, session ID, transaction ID, reward pending state, withdrawal state history, acquisition source or the Spin/Mining/Missions/Badges features exist in production. 18 such unknowns are in the assumption register; none has been converted into a fact.

What VELOOP should implement first

- A stable user_id and one user master table.
- An immutable transaction_id with idempotent wallet writes, and separate reward-versus-activity records.
- Defined value units (VE, XP, Gems, Tokens) with conversion rules, so amounts are never summed across units.
- Withdrawal state and timestamp history; UTC timestamps everywhere.
- The Critical events REQ-001 to REQ-013, validated at ingestion by the proposed rules.

Major risks

Reporting activity as earnings; summing across units; duplicate credits; unreliable daily metrics without a UTC standard; third-party outcomes outside VELOOP's control; and silently mixing simulated with production data.

Major priorities

Six Critical, seven High, five Medium and two Low backlog items (Section 19). The Critical items are all Prelaunch.

Key figures from this work (simulated data and design artefacts)

Measure	Value
Proposed events / data-dictionary fields / KPIs	38 / 100 / 46
Dashboards / visuals / developer requirements	7 / 28 / 15

Measure	Value
Quality rules designed / automated tests passing	20 / 11 of 11
Simulated datasets profiled (sheets) / records in largest	9 / 35,714 (event audit sheet)
Data quality issues found (Critical / High / Medium / Low)	3 / 11 / 15 / 5 (34 in total; see Section 12)
Referential-integrity orphans between users, sessions, events, tasks	0
Proposed KPI groups testable on supplied data (yes / partial / no)	10 / 3 / 5
Open unknowns in the assumption register	18, none converted into a fact

Recommended roadmap

- **Before launch (Critical and High):** user master and stable user_id; transaction_id and idempotent wallet writes; unit and conversion rules; reward-versus-activity separation; withdrawal state history; UTC timestamps; ingestion validation; role-based access to curated tables.
- **At launch:** emit events REQ-001 to REQ-013; run the quality checks and the reward-to-wallet reconciliation (K29) daily; certify the analytical tables that power Dashboards 1, 5 and 6.
- **After launch:** session identifier, acquisition source capture, support categories, engagement events for Spin, Missions and Streaks; remaining dashboards; replace simulated data with governed production data, never mixing them.

Decisions needed from the team

- Which of the two Users tables is the user master (assumption A016).
- Whether the behaviour workbook ('Task 02') is intended input for this task.
- Whether Rewards, Wallet, Withdrawals and Referrals datasets exist and can be supplied.
- The qualifying-activity list that defines DAU and the referral activation rule (A017, A010).